A Codebook

The 21 robotics and HRI practitioners in India discussed their own ethical decisions and six Indic frameworks.[1] Italicised quotations reproduce participant wording from the paper. Other examples paraphrase the workshop and interview findings.

Code	Description	Example
Awareness of Indic frameworks (RQ1, § 5.1)		
Standards as paperwork	Practitioners encounter industrial robot and drone safety rules in documents they must sign.	Ten recalled IS 14530 without prompting. One drone practitioner described the rules as *"only for formality purposes"*.
Ethics reduced to physical safety	Injury prevention and legal liability determine which harms count as robotics concerns.	A self-driving vehicle practitioner described attention to bodily harm while fairness, bias and dehumanisation went unexamined.
Recognition without application	Practitioners recognise a framework when prompted but have not used its provisions.	After seeing the six frameworks, 18 of 21 recognised at least one. None had applied any of them in practice.
Routes to framework awareness	Research interests and professional tasks bring different frameworks to practitioners' attention.	Practitioners encountered PPIR through Google Scholar alerts and IAIGG while preparing grant proposals.
Questioning proposed safeguards	Practitioners recall civil society proposals through disagreements about their use in robotics.	Participants questioned how to conduct IFF's impact assessments and why C-HELP opposed supervised trials that could reveal privacy risks.
Intuitive ethical practices (RQ2, § 5.2)		
Anticipating misuse	Practitioners seek criticism when planning experiments and review who may have been harmed after trials.	Seven described anticipating harm by *"accepting all manner of criticism even if it was not technical"*.
Documenting test-bed exclusions	Researchers record the deployment conditions their laboratory tests leave out.	A mapping researcher contrasted the lab's *"wide doors"* with pavements used for vending by day and sleeping by night.
Consent before mapping	Researchers explain what a mapping robot will collect and seek agreement from residents who fear surveillance.	*"residents could otherwise become features within a point cloud without having agreed to be measured in this way"*
Who must give way	Route efficiency can shift the burden of moving to people with less power over the street.	A planner questioned why a woman pushing a handcart should move aside while a parked SUV stayed put.
Being followed as a threat	People's experiences of threat change how a robot approaches or follows them.	Nurses on late shifts felt unsafe being followed. An engineer set a hospital humanoid to avoid walking behind women.
Consent and credit for demonstrators	Practitioners ask who benefits from imitation learning and whether hazardous or degrading work should be reproduced.	A sanitation project required consent, credit and a return of value for the Dalit workers and waste-pickers demonstrating tasks.
Rural households missing from simulation	Deployment in villages prompts researchers to reconsider which households their datasets represent.	After noticing which kitchens and courtyards were missing, a researcher added households from a wider range of regions and socioeconomic groups.

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[1] The six frameworks are the National Robotics Strategy (NRS), the Indian AI Governance Guidelines (IAIGG), the Postcolonial Principles for Intelligent Robotics (PPIR), Ethical Considerations of Child-Robot Interaction (ECRI), and submissions by the Internet Freedom Foundation (IFF) and the Centre for Health Equity, Law & Policy (C-HELP).

Code	Description	Example
Contextual fit of frameworks (RQ3, § 5.3)		
Refusing targeted surveillance	National guidance can give developers grounds to resist surveillance requests from state authorities.	Asked to target particular settlements, a drone developer saw the NRS as support for *"strategic refusal of invasive requests"*.
Incident records for redress	Reporting rules could encourage disclosure of robot failures without automatically penalising every incident.	Participants wanted records of robotic arm failures available to labour authorities and unions considering compensation or legal action.
Timing as part of task success	PPIR prompts a developer to reconsider a reward that treats immediate delivery as success even during prayer.	When a parcel delivery interrupted prayer, the developer had initially seen the problem *"only along the lines of odometry"*.
Residents' say over siting	Residents' knowledge of privacy and everyday movement informs where robot infrastructure can be placed.	Residents opposed a proposed drone dock because they feared the drones could look into open-air bathrooms.
Social uses of agricultural land	Practitioners use PPIR to question the assumption that fields serve only cultivation.	A couple seeking privacy in a field after dark could be misidentified by a pest-detection robot and exposed to insecticide.
Educational risks after a pilot	ECRI prompts practitioners to consider children's right to education after a school robot is withdrawn.	Some children stopped coming to class when the social robots they expected to see in lessons were withdrawn.
Perceived limitations of frameworks (RQ3, § 5.4)		
Ethics training taken for granted	Engineers struggle to interpret guidance that presumes training beyond basic safety.	Four engineers described PPIR and IAIGG as *"lacking concrete examples and implicitly assuming practitioners were already trained"*.
Caste and access to training	A practitioner links employers' reluctance to provide ethics training to caste hierarchies at work.	One practitioner described *"a false sense of ego"* among upper-caste leaders who saw engaging with minority concerns as lowering their status.
Blame assigned down the hierarchy	Practitioners read government guidance as blaming junior staff for risks authorised by their superiors.	Two participants identified *"a male-centric language of punishments and penalties"* in the NRS and IAIGG.
Consent without freedom to refuse	Recording assent cannot establish whether a person was free to refuse.	Where treatment depended on a husband's permission, one participant asked *"whose consent are we even verifying, hers or his?"*
Agency as a right to protest	Practitioners question definitions of agency that overlook a person's right to contest decisions.	Participants argued that agency includes the right to *"struggle for one's cause"*, which a definition centred on autonomy could miss.
Barriers to making frameworks work in practice (RQ3, § 5.5)		
Participation and independent review	Practitioners call for community testing and independent review of cyberphysical risks before field trials.	Participants said PPIR assumes community input, yet robots could reach trials without the involvement of people affected.
Making recruitment bias visible	Fairness depends on who demonstrates a task and who can scrutinise recruitment.	Practitioners proposed fairer demonstrator recruitment and routine third-party audits to uncover exclusions teams might miss.
Power over bodily privacy	Privacy protections depend on who can observe or exercise control over other people's bodies.	Two practitioners argued that patriarchal control over surveillance would undermine bodily privacy safeguards in the NRS and IAIGG.
Responsibility for sensor records	Data safeguards depend on explicit decisions about storing, accessing and retaining sensor records.	Participants said IAIGG assumed teams had decided who could view robot images and for how long. Such decisions were rare.